Cultivation–Depensation, Priority Effects, and Forage-Fish Harvesting in a Minimal Size-Spectrum Food Web

Size-Spectrum Ecological Modelling Study

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Abstract

The cultivation–depensation hypothesis proposes that large piscivorous fishes promote the survival of their own early life stages by cropping down smaller forage fishes that would otherwise prey upon or compete with predator larvae. We formulate a minimal three-taxa size-spectrum model in `mizer`: a semichemostat plankton resource, a small forage fish ($w_\infty = 20$ g) and a large predator ($w_\infty = 2000$ g), both spawning at 10^{-3} g. Individual variability in growth enters as size diffusion, and the equations are solved with a second-order scheme so that numerical diffusion, which in `mizer`'s default first-order scheme is as large as the physical diffusion, does not determine the results. We name each equilibrium by the components it contains (R: plankton resource; F: forage fish; P: predator). Without fishing the model has a stable coexistence state (RFP), a stable resource–forage state (RF) and an unstable interior saddle separating them; the resource–predator state (RP) is stable within its own subspace but is invaded by forage fish. Discrete invasion reproduction numbers give a priority effect: forage fish invade RP ($R_0 = 2.41$) and the community assembles to RFP, whereas predators cannot invade RF ($R_0 = 0.28$). Decomposing R_0 shows that the barrier is larval predation, not competition: forage fish in RF raise peak predation mortality on predator larvae from 1.9 to 41.5 year^{-1}, whereas the extra food they provide would on its own raise the predator's R_0. Selective fishing on the predator alone collapses it at a fold bifurcation at $F_{\mathrm{Pred}} = 0.170$ year^{-1}, after which a moratorium does not restore it. Harvesting the forage fish raises this collapse threshold. Above $F_{\mathrm{Forage}} = 0.288$ year^{-1} the predator can invade RF, but a reintroduced predator returns to the abundant state only above $F_{\mathrm{Forage}} = 0.337$ year^{-1}; in between it persists at low abundance. Mapping the $(F_{\mathrm{Forage}}, F_{\mathrm{Pred}})$ plane shows that invasion of RF leads first to a low-abundance predator state, so that the fold of the abundant state and the invasion boundary do not meet in a single point, and that the collapse of the abundant state remains abrupt until the hysteresis closes in a cusp at $F_{\mathrm{Forage}} \approx 0.650$–$0.675$ year^{-1}. The depensation is not robust to cannibalism: it disappears once forage fish prey on their own larvae at more than about 13% of the rate at which they prey on predator larvae ($\theta_{\mathrm{For,For}} > 0.120$), or once predator cannibalism exceeds $\theta_{\mathrm{Pred,Pred}} = 0.360$. Competition for the resource weakens rather than creates the depensation. When the resource regenerates slowly, the forage fish are limited by their own food in RF and the predator can invade it, although a reintroduced predator still settles at low abundance. When the resource regenerates fast, satiated forage fish eat fewer larvae. The invasion barrier is strongest at intermediate competition, close to the reference model.

1 Introduction

A long-standing puzzle in fisheries ecology is the slow or absent recovery of several piscivorous stocks after fishing was curtailed, most prominently Northwest Atlantic cod (*Gadus morhua*) (Swain and Sinclair 2000; Walters and Kitchell 2001; Frank et al. 2005). Single-species stock–recruitment theory with compensatory density dependence (Beverton and Holt 1957) predicts that a depleted

stock rebounds once fishing mortality is reduced. In several collapsed ecosystems, however, the decline of large demersal predators was accompanied by increases in small pelagic forage fishes, and the forage fishes were suspected of holding the predators down through predation on their eggs and larvae and through competition with their juveniles (Swain and Sinclair 2000; Frank et al. 2005; Swain and Mohn 2012).

Walters and Kitchell (2001) formalised this as the *cultivation–depensation* hypothesis. Abundant adult piscivores "cultivate" a favourable nursery environment by cropping down forage species; when fishing removes the adults, the forage species are released and suppress juvenile survival of the predator, producing an emergent Allee effect (depensation) at the population level. The empirical evidence is mixed. Across the North Atlantic, the abundance of small pelagics is negatively associated with cod recruitment in some stocks but not others (Minto and Worm 2012), and on the eastern Scotian Shelf a forage-fish outbreak was later followed by a collapse of the forage fish and the first signs of cod recovery (Frank et al. 2011), showing that a predator-depleted state need not be permanent. Predator–prey role reversal of this kind is also known from benthic systems (Barkai and McQuaid 1988) and has been proposed as a mechanism of hysteresis in the North Sea (Fauchald 2010).

Size-structured population theory has identified several distinct routes to such alternative stable states. In stage-structured consumer–resource models, predators that feed on one life stage of their prey can generate bistability and an emergent Allee effect through food-dependent development of the prey (de Roos and Persson 2002, 2013; van Kooten, de Roos, and Persson 2005), a mechanism that is different from cultivation–depensation but produces similar population-level signatures (Gårdmark et al. 2015; van Leeuwen, de Roos, and Persson 2008). When predator and prey each eat the other's young, the system is a form of life-history intraguild predation, whose coexistence and assembly properties depend strongly on which species is the better competitor for the shared resource (Holt and Polis 1997; Hin et al. 2011). The management implication that a predator can be restored by culling its competitors and predators of its young has been demonstrated experimentally in a whole lake (Persson et al. 2007) and explored in models of marine reserves (Baskett, Yoklavich, and Love 2006).

Multispecies size-spectrum models (Andersen and Beyer 2006; Hartvig, Andersen, and Beyer 2011; Andersen 2019) describe every fish species from egg to adult, with feeding governed by a size-based prey preference rather than by species identity. Both a 20 g forage fish and a 2 kg piscivore start life at roughly a milligram, so a predator larva necessarily passes through the size window in which adult forage fish feed, and only later grows large enough to eat them. Size-spectrum models have been used to study coexistence of size-structured species (Hartvig and Andersen 2013), the community effects of rebuilding plans (Andersen and Rice 2010) and balanced exploitation (Law, Plank, and Kolding 2012, 2016; Jacobsen, Gislason, and Andersen 2014). Here we ask whether the standard size-spectrum dynamics implemented in `mizer` (Scott, Blanchard, and Andersen 2014) produce cultivation–depensation, and if so, under which conditions. We

1. derive the discrete invasion reproduction numbers of each species at the boundary equilibria and use them to establish the assembly sequence and the priority effect;
2. decompose the predator's invasion failure into larval predation and food effects;
3. compute the fold and invasion boundaries in the plane of predator and forage-fish fishing mortality, including the intermediate states they imply;
4. vary the strength of competition for the resource, which the two species share as larvae, to ask how competition enters the mechanism; and
5. test the result against numerical resolution, the strength of size diffusion, the size range of the

resource and the strength of intraspecific predation.

2 Model Formulation and Methods

2.1 Dynamic Size-Spectrum Framework

We use the multispecies size-spectrum framework of Hartvig, Andersen, and Beyer (2011) and Andersen (2019) as implemented in `mizer` (Scott, Blanchard, and Andersen 2014) (version 3.4.0.9001). The community consists of a plankton resource spectrum $N_R(w,t)$ and two fish size spectra $N_i(w,t)$, $i \in \{\text{Forage}, \text{Predator}\}$, where $N_i(w,t)\,dw$ is the number of individuals with body mass in $[w, w+dw]$ per unit volume. All biomasses below are in grams per this (arbitrary) unit volume, which is fixed by the choice of κ.

2.1.1 Resource Dynamics

The resource spans body masses from 10^{-9} g to 1000 g and follows semichemostat dynamics in each size class:

$$\frac{\partial N_R(w,t)}{\partial t} = r_R(w)\left[c_R(w) - N_R(w,t)\right] - \mu_{p,R}(w,t)\,N_R(w,t), \tag{1}$$

with regeneration rate $r_R(w) = r_0\,w^{n-1}$ ($r_0 = 10$ g^{1-n} year^{-1}, $n = 2/3$) and predation mortality $\mu_{p,R}$ imposed by the fish. The resource spectrum at State RFP is a power law that tapers off smoothly above w_R,

$$N_R^{\mathrm{RFP}}(w) = \frac{\kappa\,w^{-\lambda}}{1 + (w/w_R)^4}, \tag{2}$$

with $\kappa = 100$, $\lambda = 2 + q - n = 2.05$ and $w_R = 10$ g. During calibration (Section 2.2) `mizer` sets the capacity to $c_R = N_R^{\mathrm{RFP}}\,(1 + \mu_{p,R}/r_R)$ so that this spectrum is a steady state. The capacity therefore coincides with N_R^{RFP} at small sizes but exceeds it by up to a factor 2.6 at $w \approx 20$ g, where the fish graze the resource most heavily. We use a smooth taper rather than a sharp upper cutoff so that the resource available to large predators declines gradually instead of ending abruptly at an arbitrary size. This is a modelling choice, not a numerical requirement: because `mizer` represents the capacity by its average over each size class, a sharp cutoff would also be integrated to second order.

2.1.2 Fish Growth, Mortality, and Reproduction

The abundance density of each fish species satisfies the McKendrick–von Foerster equation with size diffusion,

$$\frac{\partial N_i}{\partial t} + \frac{\partial (g_i\,N_i)}{\partial w} - \frac{1}{2}\frac{\partial^2 (D_i\,N_i)}{\partial w^2} = -\mu_i\,N_i, \tag{3}$$

on the size range $[w_{\min,i}, w_{\max,i}]$, with growth rate $g_i(w,t)$, size diffusion rate $D_i(w)$ and mortality rate $\mu_i(w,t)$ specified below. Predators of mass w prefer prey of mass w' according to the lognormal kernel

$$\phi_i(w/w') = \exp\left[-\frac{\left(\ln(w/(w'\,\beta_i))\right)^2}{2\,\sigma_i^2}\right], \tag{4}$$

with preferred predator–prey mass ratio β_i and width σ_i. The kernel expresses a size preference, not a strict gape limit. The available prey biomass combines resource and fish weighted by the

interaction matrix θ_{ij} (predator i, prey j):

$$E_{a,i}(w,t) = \int \phi_i(w/w') \left[N_R(w',t) + \sum_j \theta_{ij}\, N_j(w',t) \right] w'\, dw'. \tag{5}$$

With search volume $\gamma_i\, w^q$ and maximum intake rate $h_i\, w^n$ the feeding level is

$$f_i(w,t) = \frac{\gamma_i\, w^q\, E_{a,i}(w,t)}{\gamma_i\, w^q\, E_{a,i}(w,t) + h_i\, w^n}. \tag{6}$$

Assimilated energy (efficiency $\alpha = 0.6$) minus metabolic costs $k_{s,i}\, w^p$ is split between reproduction and growth:

$$E_{r,i}(w,t) = \max\Big(0,\ \alpha\, f_i(w,t)\, h_i\, w^n - k_{s,i}\, w^p\Big), \tag{7}$$

$$g_i(w,t) = E_{r,i}(w,t)\,[1 - \psi_i(w)]. \tag{8}$$

The investment function $\psi_i(w) = m_i(w)\,(w/w_{\infty,i})^{1-n}$ is the product of the maturity ogive $m_i(w) = [1 + (w/w_{\mathrm{mat},i})^{-U_i}]^{-1}$ and an allometric factor that reaches one at $w_{\infty,i}$, so that growth stops at $w_{\infty,i}$. We set the steepness $U_i = \ln 3 / \ln(w_{\mathrm{mat},i}/w_{\mathrm{mat25},i})$ by placing 25% maturity at 90% of the length at 50% maturity, $w_{\mathrm{mat25},i} = 0.9^3\, w_{\mathrm{mat},i}$, which gives $U \approx 3.5$ and a 25–75% maturation range of about 20% of the length at maturation. `mizer`'s default, $U = 10$, corresponds to a range of only 7%, steeper than typical empirical maturity ogives, and switches from immature to mature within two or three size classes of a 160-class grid.

The diffusion rate $D_i(w)$ in Eq. (3) represents the variability of growth among individuals of the same size, which arises from the randomness of prey encounters (the jump-growth term of Datta, Delius, and Law 2010) and, more importantly, from individual differences in foraging success, condition, physiology and environment. Without it, all individuals of a cohort would grow along the same trajectory. Under Eq. (3) the variance of body mass within a cohort grows at rate D_i, so the variance σ^2 of $\ln w$ increases by

$$\frac{d\sigma^2}{d\ln w} = \frac{D_i(w)}{g_i(w)\, w} \equiv c \tag{9}$$

per unit of $\ln w$ that the cohort grows. Juveniles ($\psi_i \approx 0$) feeding at the reference feeding level f_0 (Section 2.2) grow at approximately $g_i(w) \approx (\alpha f_0 h_i - k_{s,i})\, w^n$ by Eqs. (7) and (8). We therefore take $D_i(w) = D_{\mathrm{ext},i}\, w^{n+1}$ and choose $D_{\mathrm{ext},i} = c\,(\alpha f_0 h_i - k_{s,i})$, so that c is approximately constant over the juvenile size range. In the reference model $c = 0.10$: a cohort that grows from egg (10^{-3} g) to the predator's maturation size spans $\ln(250/10^{-3}) \approx 12.4$ units of $\ln w$ and acquires a standard deviation of $\ln w$ of about 1.1, i.e. adults of the same age differ in mass by a typical factor of 3.0. Because this level is uncertain, we report all thresholds for c between 0.02 and 0.3 (Section 3.7). Mean growth stops at $w_{\infty,i}$, but size diffusion carries some individuals beyond it; a senescence mortality (below) keeps them from spreading further.

Total mortality $\mu_i(w,t) = \mu_{p,i}(w,t) + \mu_{\mathrm{ext},i}(w) + F_i\, S_i(w)$ comprises:

1. **Predation mortality** imposed by all fish species j:

$$\mu_{p,i}(w',t) = \sum_j \theta_{ji} \int \phi_j(w/w')\,[1 - f_j(w,t)]\,\gamma_j\, w^q\, N_j(w,t)\, dw; \tag{10}$$

2. **External mortality** $\mu_{\mathrm{ext},i}(w) = z_{0,i} + z_{\mathrm{ext},i}\, w^{\,n-1} + z_{\mathrm{sen}}\,(w/w_{\infty,i})^{k}$ with w in grams. Because $n-1 = -1/3$, the second term represents mortality from unmodelled predators of small fish; it is largest for larvae (5.3 and 12.1 year^{-1} at $w_{\min}$ for forage fish and predator) and small for adults. The third term ($z_{\mathrm{sen}} = 0.5$ year^{-1}, $k = 10$) is a senescence mortality that is negligible below $0.8\,w_{\infty,i}$ and rises steeply above $w_{\infty,i}$. It is needed because size diffusion continues where mean growth stops: without it, adults would random-walk to several times $w_{\infty,i}$, where only the small background mortality removes them, and the results would depend on where the size grid ends. With it, 3% of predator biomass at State RFP lies above w_{∞}, and the grid, which ends at $w_{\max,i} = 2\,w_{\infty,i}$, holds a negligible density at its upper end;
3. **Fishing mortality** $F_i\, S_i(w)$ with logistic length-based selectivity

$$S_i(w) = \left[1 + \exp\!\left(\ln 3\,\frac{l_{50,i} - l(w)}{l_{50,i} - l_{25,i}}\right)\right]^{-1}, \qquad l(w) = (w/a)^{1/b}, \tag{11}$$

where $l_{50,i}$ and $l_{25,i}$ are the lengths at 50% and 25% selection and $w = a\,l^{b}$ with $a = 0.01$ g cm^{-3} and $b = 3$. F_i is thus the fishing mortality on fully selected fish.

The density-independent egg production rate is

$$R_{\mathrm{DI},i}(t) = \frac{1}{2}\,\frac{\epsilon_i}{w_{\min,i}} \int_{w_{\min,i}}^{w_{\max,i}} E_{r,i}(w,t)\,\psi_i(w)\,N_i(w,t)\,dw, \tag{12}$$

where the reproductive efficiency ϵ_i (`erepro` in `mizer`) also absorbs egg and early larval losses not otherwise modelled. Recruitment follows a Beverton–Holt relationship (Beverton and Holt 1957)

$$R_{\mathrm{DD},i}(t) = \frac{R_{\mathrm{DI},i}(t)}{1 + R_{\mathrm{DI},i}(t)/R_{\max,i}}, \tag{13}$$

which sets the flux $g_i N_i$ at $w_{\min,i}$. We parameterise the compensation by the reproduction level $L_{R,i} = R^{*}_{\mathrm{DD},i}/R_{\max,i}$ at State RFP.

2.2 Community Parameterization

Table 1 lists the parameters of the model as used in the simulations (`build_cultivation_model()` in `cultivation_depensation.R`). Both species recruit at $w_{\min} = 10^{-3}$ g. Adult forage fish ($w \in [4, 20]$ g, $\beta = 200$) feed preferentially on prey of 0.02–0.1 g, the larval and early juvenile sizes of the predator. Maturing and adult predators ($w \in [250, 2000]$ g, $\beta = 100$) prefer prey of 2.5–20 g, the juvenile and adult forage fish, but also take the larger resource classes around $w_R = 10$ g. Maximum intake rates h_i follow from the von Bertalanffy growth rates, search volumes γ_i from the reference feeding level $f_0 = 0.6$, and metabolic coefficients $k_{s,i}$ from `mizer`'s default critical feeding level of 0.2 (Andersen 2019).

The interaction matrix is

$$\boldsymbol{\theta} = \begin{pmatrix} \theta_{\mathrm{For,For}} & \theta_{\mathrm{For,Pred}} \\ \theta_{\mathrm{Pred,For}} & \theta_{\mathrm{Pred,Pred}} \end{pmatrix} = \begin{pmatrix} 0.05 & 0.95 \\ 1.00 & 0.05 \end{pmatrix}. \tag{14}$$

This matrix is strongly asymmetric. Adult forage fish encounter predator larvae at almost full rate ($\theta_{\mathrm{For,Pred}} = 0.95$) but their own larvae, which have exactly the same size, at only 5% of that rate. Biologically this corresponds to spatial or temporal separation of forage-fish spawning from forage-fish feeding grounds, combined with overlap between forage-fish adults and predator spawning

and nursery areas, as proposed for pelagic fishes feeding on cod eggs and larvae (Swain and Sinclair 2000; Fauchald 2010). Predator cannibalism is likewise weak ($\theta_{\mathrm{Pred,Pred}} = 0.05$). Section 3.7 shows that both choices are essential for the results.

The reference model is defined by a fixed calibration recipe, not by the calibration targets, which are illustrative rather than observations. With nominal State RFP biomasses $B_{\mathrm{Forage}} = 100$ g and $B_{\mathrm{Predator}} = 600$ g as `biomass_observed`, `build_cultivation_model()` applies `steadySingleSpecies()` and then two rounds of `matchBiomasses()` followed by `tuneSteadyState()` (projection solver, $\Delta t = 0.2$ year, default tolerances), with recruitment still density independent. This determines ϵ_i and the resource capacity. It then sets the reproduction levels to $L_R = 0.65$ (forage) and 0.78 (predator) with `reproduction_level<-`, which fixes $R_{\max,i}$ and rescales ϵ_i, and solves for the fixed point with `findSteadyState(solver = "newton")`, which changes no parameter. Two rounds do not converge to the targets: the resulting State RFP holds 93.8 g of forage fish and 584.3 g of predators, and because the final Newton solve moves the state slightly, the reproduction levels are 0.6511 and 0.7801. We accept this calibration error and take the resulting model, saved as `cultivation_depensation_params.rds`, as the reference model: every number in this paper refers to it, not to the nominal targets. A rerun of the recipe reproduces it exactly. Calibrating instead to the nominal targets themselves changes the thresholds quantitatively (Section 3.7). Two separate gears with unit catchability, so that effort equals the fishing mortality on fully selected fish, act on forage fish with 50% selection at 1 g ($l_{50} = 4.6$ cm; juveniles and adults, since maturation is at 4 g) with mortality F_{Forage}, and on predators with 50% selection at 400 g ($l_{50} = 34.2$ cm) with mortality F_{Pred}. For both gears $l_{25} = 0.9\, l_{50}$, a selection range ($l_{75} - l_{25}$) of 20% of l_{50}, typical of trawl gear; the selectivity rises from 25% to 75% over about 0.6 units of $\ln w$, i.e. over many size classes. The choice of a logistic rather than a knife-edge selectivity is biological, not numerical: because `mizer` averages the selectivity over each size class, the second-order scheme would handle a knife edge with second-order accuracy as well. The size axis is discretised into 320 logarithmically spaced classes ($\Delta \ln w = 0.048$). We use `mizer`'s second-order finite-volume scheme (`second_order_w`), with a centred advective flux and bin-averaged rates, so that the numerical diffusion is small compared with the physical diffusion. This matters: `mizer`'s default first-order upwind scheme adds numerical diffusion equivalent to $c \approx \Delta \ln w$, which on a grid of 80 classes ($c \approx 0.19$) exceeds the physical diffusion assumed here and has the wrong dependence on the grid (Section 3.7). The resource grid extends down to 10^{-9} g, so that the feeding kernels of newly hatched larvae, which prefer prey of 5×10^{-6} g (forage fish) and 10^{-5} g (predator), lie on the grid to more than six standard deviations below their peak. Time-dependent simulations use the second-order TR-BDF2 method with $\Delta t = 0.1$ year; steady states are solved for with `mizer`'s Newton solver.

2.3 Steady States, Stability, and Invasion Reproduction Numbers

We name each equilibrium by the components it contains: R for the plankton resource, F for the forage fish and P for the predator. Starting from the calibrated coexistence equilibrium (**State RFP**) we obtain the boundary equilibria with *identical* parameters (`get_fixed_points()`): **State RP** (resource + predator) by removing the forage fish, projecting and solving for the fixed point, **State RF** (resource + forage fish) by removing the predator, and **State R** (resource only) with both fish absent and $N_R = c_R$. We judge convergence by the drift of each species' total biomass over a subsequent 100-year projection (TR-BDF2, $\Delta t = 0.1$ year), which is less than a fraction 2×10^{-9} per year in all four states. We do not use `mizer`'s `getSteadyResidual()` for this: it reports rates per size class, which with size diffusion are dominated by nearly empty classes at the upper end of

Table 1: Parameters of the two fish species as used in the simulations. Resource: $\kappa = 100$, $\lambda = 2.05$, $r_0 = 10$ g$^{1/3}$ year^{-1}, size range 10^{-9}–1000 g, taper at $w_R = 10$ g. Interaction matrix in Eq. eqrefeq:inter.

Parameter	Forage	Predator
Asymptotic mass w_∞ [g]	20	2000
Largest size class $w_{\max}$ [g]	40	4000
Maturation mass w_{mat} [g]	4	250
Mass at which a quarter are mature, w_{mat25} [g]	2.92	182
Egg mass $w_{\min}$ [g]	0.001	0.001
Preferred predator–prey mass ratio β	200	100
Feeding kernel width σ	1.3	1.5
von Bertalanffy growth rate k_{vb} [year^{-1}]	1	0.5
Reference feeding level f_0	0.6	0.6
Search volume coefficient γ	0.0745	0.177
Maximum intake coefficient h [g^{1-n} year^{-1}]	21.2	55.9
Metabolic coefficient k_s [g^{1-p} year^{-1}]	2.42	5.58
Exponents n, p, q	0.667, 0.70, 0.717	0.667, 0.70, 0.717
Assimilation efficiency α	0.6	0.6
External mortality z_0 [year^{-1}]	0.3	0.2
External mortality z_{ext} [g$^{1/3}$ year^{-1}]	0.5	1.2
Reproductive efficiency ϵ (calibrated)	0.093	3.03e-05
Maximum recruitment $R_{\max}$ (calibrated)	823	1.78
Reproduction level L_R at State RFP	0.651	0.78
Size diffusion coefficient D_{ext} [g$^{1/3}$ year^{-1}]	0.519	1.45
Length at which half the fish are selected, l_{50} [cm]	4.64	34.2
Length at which a quarter are selected, l_{25} [cm]	4.18	30.8

the size range, and it includes a flux into the last size class of the predator's grid, a class that the simulation holds at zero.

Invasion reproduction number. For a species i absent from a resident equilibrium, the resident environment fixes its growth rate $g_i(w)$, diffusion $D_i(w)$, mortality $\mu_i(w)$ and reproductive investment. We solve the linear steady-state transport equation of the discretised model for a unit density in the first size class, compute the resulting lifetime egg production $R^{(1)}_{\mathrm{DI},i}$ with Eq. (12), and divide by the recruitment flux $R_{\mathrm{req},i}$ needed to sustain that density, which we read from the balance of the first size class in the same discretisation:

$$R_{0,i} = \frac{R^{(1)}_{\mathrm{DI},i}}{R_{\mathrm{req},i}}. \tag{15}$$

Because the Beverton–Holt function has unit slope at zero, a rare species increases if and only if $R_{0,i} > 1$ (Diekmann, Heesterbeek, and Metz 1990); we confirm this below against the leading eigenvalue. For a resident species the same quantity equals $R^*_{\mathrm{DI}}/R^*_{\mathrm{DD}} = 1/(1 - L_R) > 1$; it measures the compensatory reserve of recruitment rather than invasion. We use this identity as a check of the calculation, which it passes to four significant figures. Section 3.7 reports how R_0 changes with the number of size classes.

Linear stability. We compute the eigenvalues of the Jacobian of the discretised fish–resource system with `getStability()`. `getStability()` linearises only over species that are present; to obtain the stability of a boundary equilibrium in the full system, we add the absent species at a trace density (10^{-12} of its State RFP abundance), which adds its invasion exponent to the spectrum. We confirm the invasion exponent against the decay of a predator seed in direct simulation.

Unstable interior equilibrium. The saddle that separates the basins of attraction of States RFP and RF is located by edge tracking: we bisect on the size of a predator seed added to State RF to find the basin boundary, follow a trajectory started on it to where it lingers, and refine that state with `mizer`'s Newton solver.

Bifurcation boundaries. For each forage-fish mortality $F_{\mathrm{Forage}} \in \{0, 0.025, \ldots, 0.7\}$ year^{-1} we compute (i) the predator invasion boundary F_{inv}, where $R_{0,\mathrm{Pred}}(\text{State RF}) = 1$; (ii) the upper limit F_{up} of the abundant coexistence state, by continuation from State RFP in steps of 0.01 followed by bisection to 0.001, classifying a state as abundant when the predator biomass exceeds 100 g; and (iii) the lower limit F_{low} of the depleted state, by continuation downward from State RF with a predator seed. Each step is projected (with the first-order time step, $\Delta t = 0.5$ year, which leaves the fixed points unchanged) until every biomass changes by less than 10^{-5} per year; this strict criterion is needed because trajectories linger near a fold. Finally, we determine by bisection the smallest forage-fish mortality $F_{\mathrm{Forage,rec}}$ at which a predator reintroduced into State RF (at 0.5% of its State RFP abundance, with $F_{\mathrm{Pred}} = 0$) returns to the abundant state. A leading eigenvalue that tends to zero as $F_{\mathrm{Pred}} \to F_{\mathrm{up}}$ identifies F_{up} as a fold (saddle-node) bifurcation. Where $F_{\mathrm{low}} < F_{\mathrm{inv}}$, the depleted state below F_{inv} contains a small predator population, so the invasion bifurcation is supercritical.

Robustness. We rebuild and recalibrate the model with half and twice as many size classes; with the first-order upwind scheme, 80 size classes and no physical diffusion (the discretisation of the first version of this study); with the resource taper at $w_R = 1$ g instead of 10 g; for size diffusion $c \in \{0.02, 0.05, 0.1, 0.2, 0.3\}$; and over a range of forage-fish and predator cannibalism ($\theta_{\mathrm{For,For}}$ and $\theta_{\mathrm{Pred,Pred}}$ from 0.05 to 0.95, one at a time, all other entries of $\boldsymbol{\theta}$ unchanged). For $c < \Delta \ln w$ (here only $c = 0.02$) the centred flux is no longer monotone and we use the van Leer limiter instead. So that each variant differs from the reference model only in the property being varied, every variant

is first built with the recipe above and then calibrated further (`recalibrate()`) until its State RFP biomasses equal those of the reference model: we alternate `matchBiomasses()` with Newton solves of `tuneSteadyState()` until the largest relative biomass error is below 10^{-6}, impose the reproduction levels of the reference model and solve for the fixed point without changing parameters, and check that biomasses and reproduction levels then differ from the reference values by less than 10^{-5}. Each variant thus has its own ϵ_i, $R_{\max,i}$ and resource capacity (all saved with the variant). One further variant instead calibrates the reference model in the same way to the nominal targets (100 g, 600 g; $L_R = 0.65, 0.78$). The interaction strengths at which $R_{0,\mathrm{Pred}}(\mathrm{RF}) = 1$ are located by root finding (tolerance 10^{-4}) on separately calibrated models.

Competition for the resource. The fish compete for the resource only through its depletion, and how strongly the resource responds to a change in grazing is set by its regeneration rate. At a steady state of Eq. (1) the resource level is

$$\frac{N_R}{c_R} = \frac{r_R}{r_R + \mu_{p,R}}, \qquad \frac{\partial \ln N_R}{\partial \mu_{p,R}} = -\frac{1}{r_R + \mu_{p,R}}, \tag{16}$$

so for $r_R \ll \mu_{p,R}$ the fish hold the resource far below its capacity and any change in their abundance changes the food available to all other consumers, whereas for $r_R \gg \mu_{p,R}$ the resource stays at its capacity whatever the fish do, and there is no exploitative competition. We vary the coefficient r_0 of the regeneration rate from 0.25 to 1000 $\mathrm{g}^{1/3}$ year^{-1} (reference value 10) and set the capacity each time to $c_R = N_R^{\mathrm{RFP}}\,(1 + \mu_{p,R}/r_R)$ (`resource_rate<-` with `balance = TRUE`). State RFP, with its resource spectrum and every fish parameter, is therefore identical in all variants, and no recalibration is needed: the variants differ only in how the resource, and through it the fish, respond to a departure from State RFP. For each r_0 we compute States RP and RF, the invasion numbers, the decomposition of $R_{0,\mathrm{Pred}}(\mathrm{RF})$ into growth and mortality, and the thresholds F_{crit}, $F_{\mathrm{Forage,inv}}$ and $F_{\mathrm{Forage,rec}}$ as above, and locate the values of r_0 at which $R_{0,\mathrm{Pred}}(\mathrm{RF}) = 1$ by root finding on $\ln r_0$.

3 Results

3.1 Unfished Alternative Stable States and the Depensatory Bottleneck

Without fishing the model has the four boundary and interior equilibria listed in Table 2 and, in addition, an unstable interior saddle. States RFP and RF are locally asymptotically stable in the full system. State RP is stable within the resource–predator subspace but is a saddle in the full system, because forage fish invade it. State R is invaded by both fish species.

The saddle lies at $B_{\mathrm{Forage}} = 781$ g and $B_{\mathrm{Predator}} = 290$ g and has a single unstable eigenvalue (0.206 year^{-1}). The predator invasion exponent at State RF, $\lambda_{\mathrm{full}} = -0.291$ year^{-1}, agrees with the decay rate of a predator seed in direct simulation (-0.290 year^{-1}; Figure 2B), and the forage-fish invasion exponent at State RP is 0.99 year^{-1}, confirming the R_0 criteria.

Comparing States RFP and RF exposes the mechanism (Figure 1). In State RFP the predator ($B^*_{\mathrm{Pred}} = 584.3$ g) holds forage-fish biomass down to 93.8 g, and predation mortality on predator larvae ($w \le 1$ g) is at most 1.9 year^{-1}. Without the predator, forage-fish biomass rises 23-fold to 2169 g, grazes a trough into the resource around 0.01–0.5 g and raises predation mortality on predator larvae to a peak of 41.5 year^{-1} at $w \approx 0.04$ g. Survival of a predator from egg to maturation along the mean growth trajectory, $\exp\big(-\int_{w_{\min}}^{w_{\mathrm{mat}}} \mu_{\mathrm{Pred}}(w)/g_{\mathrm{Pred}}(w)\,dw\big)$ (evaluated as a sum over the size classes whose lower edge does not exceed w_{mat}, and ignoring size diffusion), falls from 0.2 in State RFP to 0.0041 in State RF, a factor of 49.

Table 2: The four boundary and interior equilibria without fishing. R_0 of an absent species is its invasion reproduction number; for a resident species it equals $1/(1 - L_R)$. λ_{res} is the leading eigenvalue [year^{-1}] within the resident subspace (for State R, the slowest resource relaxation rate), λ_{full} the leading eigenvalue including the invasion exponents of absent species. Eigenvalues of unreachable size classes are excluded. Stable: $\lambda_{\text{full}} < 0$.

Resident taxa	B^*_{For} [g]	B^*_{Pred} [g]	$R_{0,\text{For}}$	$R_{0,\text{Pred}}$	λ_{res}	λ_{full}	Stable
Resource only	0.0	0.0	459.05	9.18	-1.006	4.182	No
Resource + Forage + Predator	93.8	584.3	2.87	4.55	-0.311	-0.311	Yes
Resource + Predator	0.0	537.6	2.41	3.61	-0.625	0.987	No
Resource + Forage	2169.0	0.0	111.17	0.28	-0.625	-0.291	Yes

The forage fish affect the predator in two ways: they eat its larvae, and they compete with its larvae for the resource while providing food for its adults. To separate the two, we compute $R_{0,\text{Pred}}$ by Eq. (15) with the predator's growth rate g_{Pred} and reproductive investment $E_{r,\text{Pred}}\psi_{\text{Pred}}$ from one state and its total mortality μ_{Pred} from the other (size diffusion does not depend on the state). With State RFP growth and State RF mortality, $R_{0,\text{Pred}} = 0.136$; with State RF growth and State RFP mortality, $R_{0,\text{Pred}} = 11.8$. Without substitution the same calculation returns 4.547 in State RFP, equal to $1/(1 - L_R)$, and 0.276 in State RF, as in Table 2. The invasion barrier is therefore caused entirely by larval predation. The food effect acts in the opposite direction: in State RF, adult predators would find abundant forage fish and grow and reproduce faster than in State RFP.

Even in State RFP, adult predators take only 20% of their food from forage fish and 80% from the resource, which extends beyond 10 g. In this model the "predator" is a generalist whose population does not depend on the forage fish for food, which is also why it persists in State RP.

3.2 Priority Effects and Sequential Community Assembly

The invasion reproduction numbers define the assembly sequence from the resource-only State R:

1. Both fish species can invade State R ($R_{0,\text{For}} = 459$, $R_{0,\text{Pred}} = 9.2$).
2. If the predator establishes first (State R $\to$ RP), forage fish can invade ($R_{0,\text{For}} = 2.41$). A forage seed of 10^{-3} of its State RFP abundance grows and the community converges to State RFP (Figure 2A), with predator biomass increasing from 537.6 to 584.3 g.
3. If the forage fish establishes first (State R $\to$ RF), the predator cannot invade ($R_{0,\text{Pred}} = 0.28$). A predator seed of 10^{-2} of its State RFP abundance briefly increases as the seeded adults spawn, then declines exponentially at a rate of 0.290 year^{-1} (Figure 2B).

Without fishing the system is therefore bistable between States RFP and RF, and assembly of the full food web from rare invaders requires the predator to arrive first. Because the forage fish has the much larger invasion number in State R, simultaneous arrival of rare colonists of both species would favour State RF. Priority effects of this kind are expected in life-history intraguild predation, where each species preys on the other's young (Holt and Polis 1997; Hin et al. 2011).

3.3 Selective Predator Exploitation and Catastrophic Collapse

When only the predator is fished, increasing F_{Pred} gradually lowers predator biomass and releases the forage fish (Figure 3, left), until the abundant coexistence state disappears at $F_{\text{crit}} = 0.170$ year^{-1}.

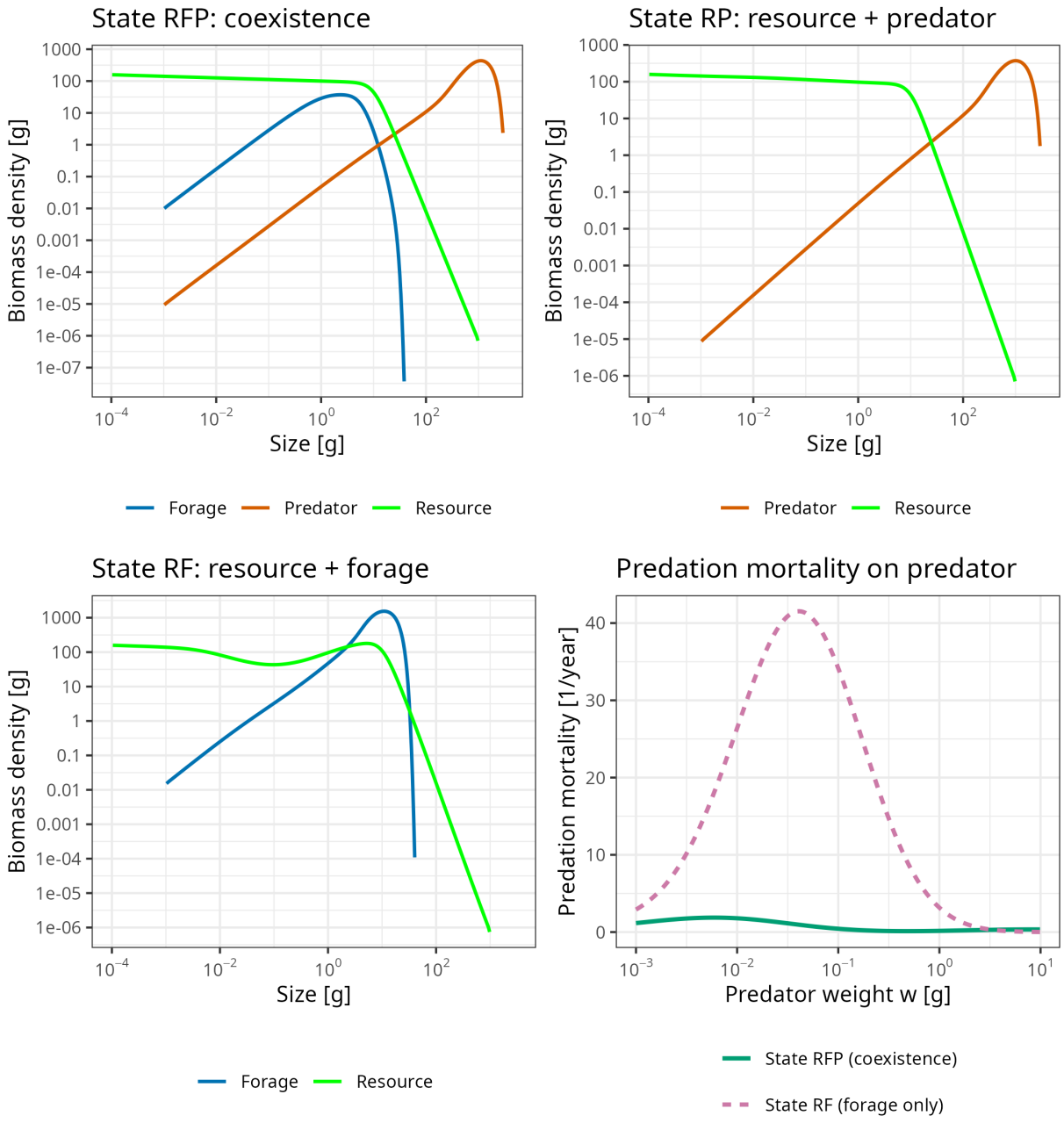


Figure 1: Unfished equilibrium size spectra and the depensatory bottleneck. Top left: coexistence State RFP. Top right: resource–predator State RP. Bottom left: resource–forage State RF, where release from predation allows the forage fish to expand and graze down the resource. Bottom right: predation mortality on predator early life stages ($w \leq 10$ g) in State RFP (solid green) and State RF (dashed magenta).

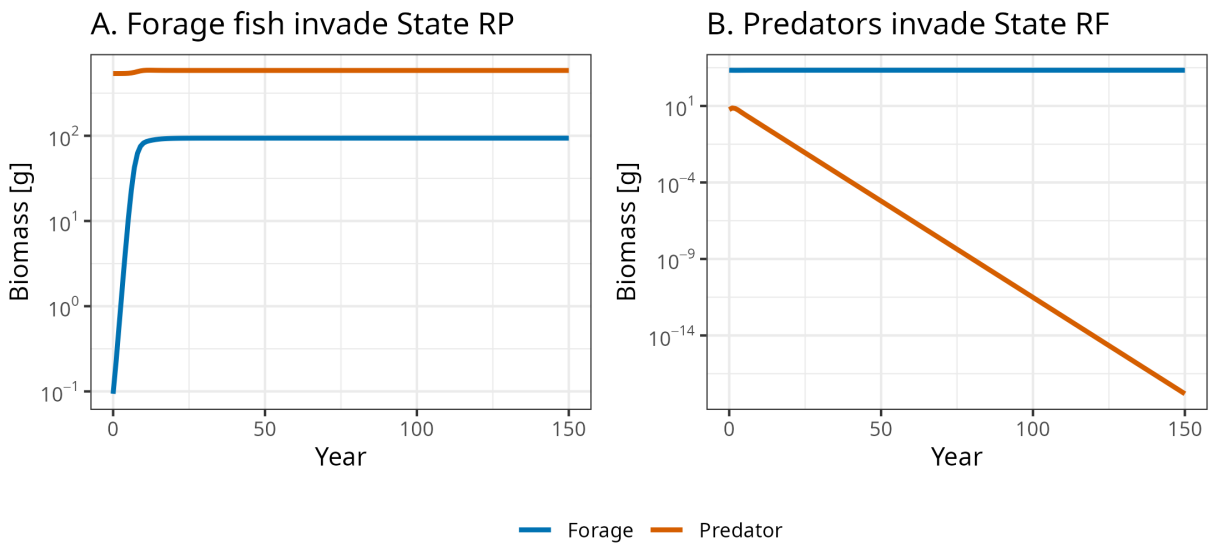


Figure 2: Invasion experiments without fishing. (A) A trace population of forage fish introduced into the resource–predator State RP invades and the community converges to State RFP. (B) A seed population of predators introduced into the resource–forage State RF declines exponentially, because predation by forage fish on predator larvae keeps $R_{0,\mathrm{Pred}} < 1$.

Just below this threshold the predator biomass is still 392 g, and the leading eigenvalue is within 0.018 year^{-1} of zero, as it is at every point of the upper boundary below the cusp (Section 3.5): State RFP is lost at a fold bifurcation, where it merges with the saddle. Above F_{crit} the community jumps to State RF. Because State RF is uninvasible for every $F_{\mathrm{Pred}} \geq 0$ when $F_{\mathrm{Forage}} = 0$, reducing F_{Pred} again does not bring the predator back: the decreasing sweep stays in State RF down to $F_{\mathrm{Pred}} = 0$.

Harvesting the forage fish at $F_{\mathrm{Forage}} = 0.35$ year^{-1} (Figure 3, right) raises the collapse threshold to $F_{\mathrm{up}} = 0.253$ year^{-1}, again a fold (predator biomass 321 g just below it). The state it collapses to is different. At this forage harvest the predator can invade State RF for $F_{\mathrm{Pred}} < F_{\mathrm{inv}} = 0.094$ year^{-1}, and the invasion leads to a low-abundance coexistence state rather than to State RFP. When fishing pressure is reduced again, this low-predator state persists down to $F_{\mathrm{Pred}} = 0.016$ year^{-1} before the predator recovers to the abundant state, so the hysteresis loop spans 0.016–0.253 year^{-1}.

3.4 Hysteresis Under Moratorium and Recovery via Forage-Fish Harvesting

Figure 4 follows a five-phase management scenario:

1. **Years 0–50, unfished**: the community is at State RFP.
2. **Years 50–200, predator overfishing** ($F_{\mathrm{Pred}} = 0.25$ year^{-1}, above F_{crit}; $F_{\mathrm{Forage}} = 0$): the predator collapses and forage fish rise to 2169 g.
3. **Years 200–320, moratorium**: all fishing stops and predators are reintroduced at 0.5% of their State RFP abundance. Because $R_{0,\mathrm{Pred}}$(State RF) = 0.28, the reintroduced population declines.
4. **Years 320–470, forage-fish harvest**: forage fish are fished at $F_{\mathrm{Forage}} = 0.40$ year^{-1} and predators are reintroduced again. This is above $F_{\mathrm{Forage,rec}} = 0.337$ year^{-1} (Section 3.5).

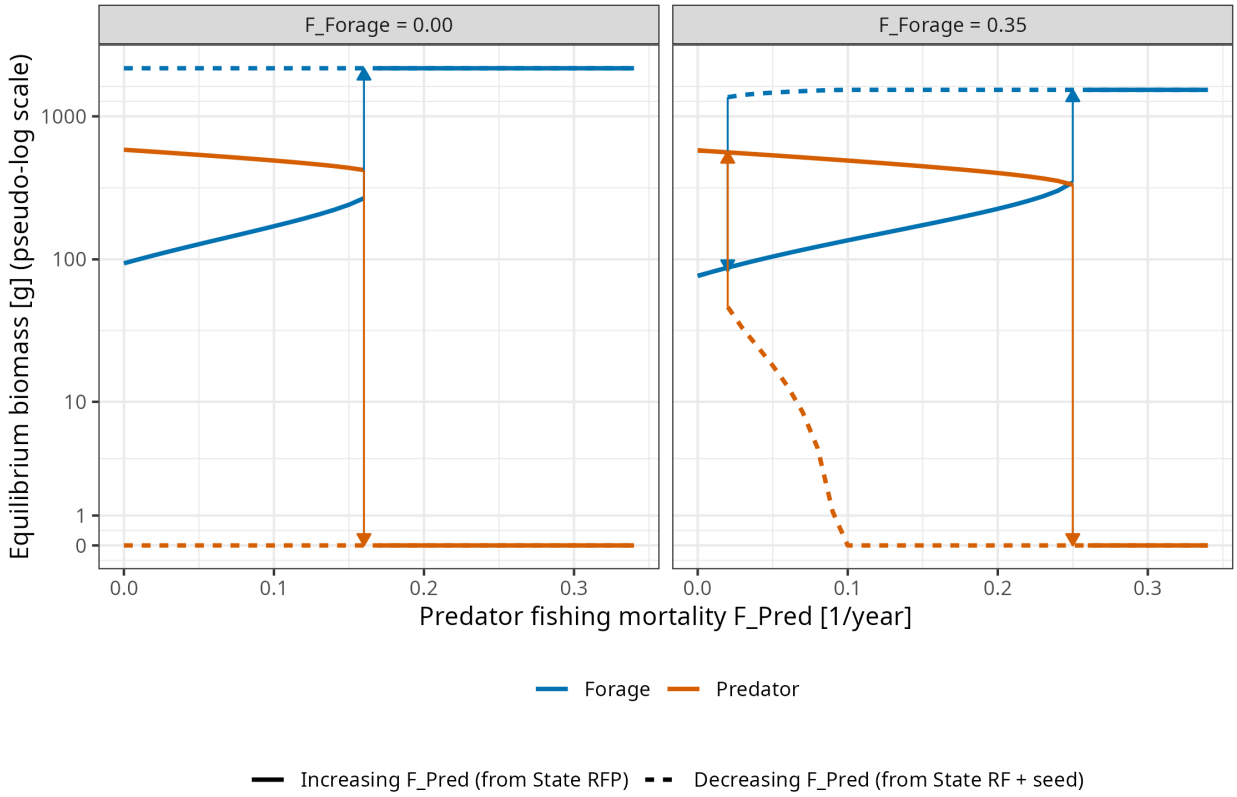


Figure 3: Equilibrium biomass of forage fish (blue) and predator (orange) along increasing (solid, starting from State RFP) and decreasing (dashed, starting from State RF with a predator seed) predator fishing mortality, without forage-fish harvesting (left) and with $F_{\text{Forage}} = 0.35$ year^{-1} (right). Thin arrows mark where a branch of stable states ends and the system jumps to another. The vertical scale is pseudo-logarithmic so that both zero and small predator biomasses are visible. Where the two sweeps differ, the system is bistable.

Forage-fish biomass in State RF falls to 1464 g and $R_{0,\mathrm{Pred}}$ rises to 1.41. The predator invades, exceeds half its State RFP biomass by year 397, and suppresses the forage fish.

5. **Years 470–570, unfished**: when the forage fishery stops, the community remains in State RFP ($B_{\mathrm{Pred}} = 584.3$ g, $B_{\mathrm{For}} = 93.8$ g), the state it occupied before overfishing.

The deterministic model can only recover the predator from a reintroduction or immigration, since an extinct population cannot invade; this is why predators are seeded in phases 3 and 4.

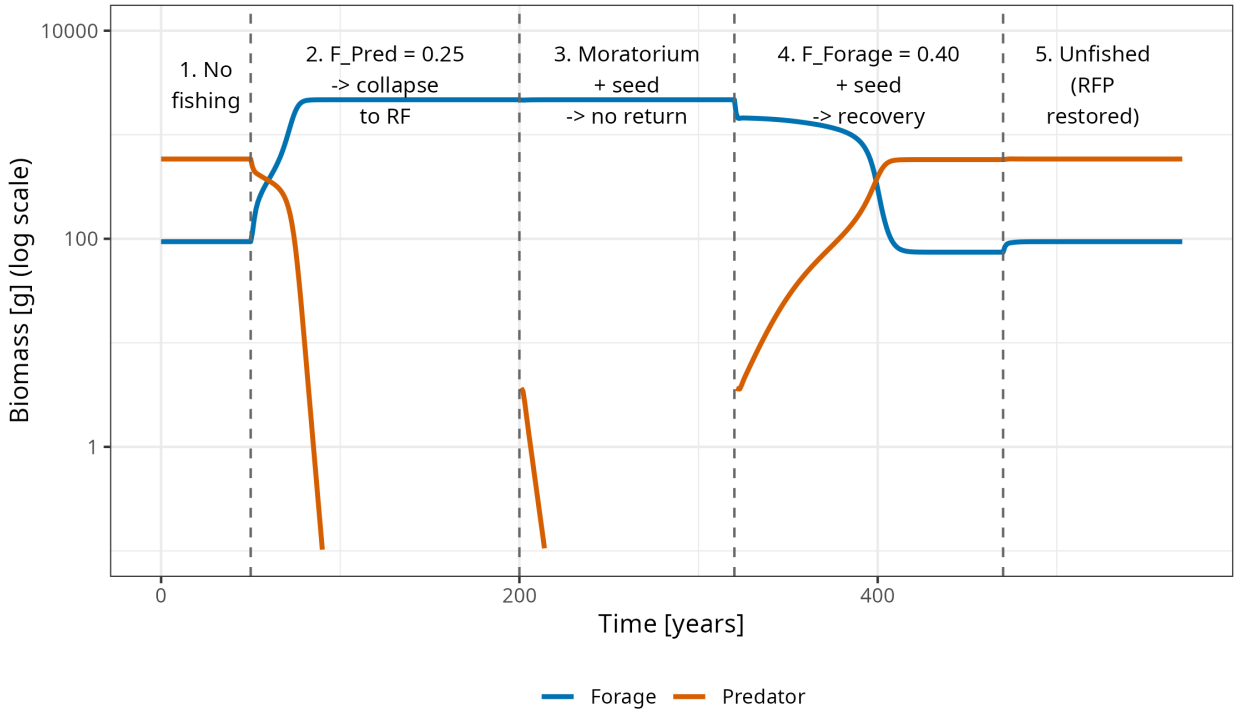


Figure 4: Five-phase management scenario. Phase 2: selective predator overfishing collapses the predator. Phase 3: a moratorium with predator reintroduction fails because State RF is uninvasible. Phase 4: harvesting the forage fish makes State RF invasible and the reintroduced predator recovers. Phase 5: after all fishing stops, the restored State RFP persists.

3.5 Two-Parameter Exploitation Diagram

In State RF, harvesting the forage fish reduces its biomass and increases $R_{0,\mathrm{Pred}}$ (Figure 5A), from 0.28 at $F_{\mathrm{Forage}} = 0$ to 2.72 at $F_{\mathrm{Forage}} = 0.7$ year^{-1}. The predator can invade State RF once

$$F_{\mathrm{Forage}} > F_{\mathrm{Forage,inv}} = 0.288 \text{ year}^{-1}, \tag{17}$$

at which State RF holds 1623 g of forage fish.

Figure 5B shows the computed regimes in the $(F_{\mathrm{Forage}}, F_{\mathrm{Pred}})$ plane. The fold that bounds the abundant coexistence state, F_{up}, rises from 0.170 year^{-1} without forage harvest to 0.226 at $F_{\mathrm{Forage}} = 0.25$ and 0.253 at $F_{\mathrm{Forage}} = 0.35$ year^{-1}. The predator invasion boundary of State RF, F_{inv}, rises more steeply, from zero at $F_{\mathrm{Forage,inv}}$ to 0.094 year^{-1} at $F_{\mathrm{Forage}} = 0.35$. The two curves cross, but they do not meet at a single codimension-two point as the simplest picture would suggest. Instead the plane divides as follows:

1. **No invasion** ($F_{\text{Forage}} < F_{\text{Forage,inv}}$). State RF is uninvasible for every $F_{\text{Pred}} \geq 0$, so the abundant state and State RF are both stable from $F_{\text{Pred}} = 0$ up to the fold F_{up} (amber region), and only State RF remains above it. This is why a moratorium on predator fishing alone cannot restore the predator.
2. **Invasion to a low-predator state** ($F_{\text{Forage}} > F_{\text{Forage,inv}}$). Wherever $F_{\text{inv}} > 0$, the invasion bifurcation of State RF is supercritical: a predator invading at F_{Pred} just below F_{inv} settles at a low abundance rather than returning to the abundant state. The low-predator state persists down to its own lower limit F_{low}, below which the predator recovers fully. Between $\max(0, F_{\text{low}})$ and F_{up} the abundant state is bistable with State RF (amber, above F_{inv}) or with the low-predator state (yellow, below F_{inv}), and where $F_{\text{up}} < F_{\text{inv}}$ the low-predator state is the only attractor between them (pink). Because F_{low} is negative only for $F_{\text{Forage}} > F_{\text{Forage,rec}} = 0.337$ year^{-1}, a predator reintroduced without any predator fishing recovers fully only above this forage harvest; for $0.288 < F_{\text{Forage}} < 0.337$ year^{-1} it persists only as a remnant.
3. **Cusp** ($F_{\text{Forage}} \approx 0.650$–$0.675$ year^{-1}). The upper fold of the abundant state and the lower limit of the low-predator state approach each other as forage harvest increases: the width of the bistable interval shrinks from 0.237 year^{-1} at $F_{\text{Forage}} = 0.35$ to 0.036 year^{-1} at $F_{\text{Forage}} = 0.6$ and has closed at 0.675 year^{-1}. Beyond the cusp the predator declines continuously with increasing F_{Pred} and recovers along the same path when fishing is reduced.

Forage-fish harvesting therefore removes the regime shift only at forage-fish fishing mortalities above about 0.65 year^{-1}, well beyond the level needed to make State RF invasible.

3.6 Competition for the Resource

Changing the resource regeneration rate leaves State RFP unchanged but changes how strongly the fish deplete the resource: the lowest resource level N_R/c_R at State RFP over the sizes 10^{-6}–10 g eaten by fish larvae and juveniles is 0.017 at $r_0 = 0.25$, 0.40 in the reference model and 0.985 at $r_0 = 1000$. In State RF the strength of competition matters a great deal (Figure 6). Without the predator, the forage fish are limited by their own food when the resource regenerates slowly: State RF holds 547 g of forage fish at $r_0 = 0.25$, 2169 g in the reference model and 2759 g at $r_0 = 1000$. The predator's invasion number does not follow this monotonic trend. $R_{0,\text{Pred}}(\text{RF})$ falls from 1.68 at $r_0 = 0.25$ to a minimum of 0.250 at $r_0 = 5$, close to the reference value, and rises again to 0.554 at $r_0 = 1000$. It exceeds one, so that the predator can invade State RF without any fishing, only for $r_0 < 0.595$ g$^{1/3}$ year^{-1}, where the fish hold the resource at less than about 4% of its capacity at the most heavily grazed sizes.

The decomposition of $R_{0,\text{Pred}}(\text{RF})$ into mortality and growth (Section 3.1) explains this pattern (Figure 6A):

1. *Larval predation is strongest at intermediate competition.* With State RFP growth and State RF mortality, $R_{0,\text{Pred}}$ falls from 2.07 at $r_0 = 0.25$ to 0.136 at $r_0 = 10$ as the forage fish become more abundant, and peak predation mortality on predator larvae rises from 17.5 to 41.5 year^{-1}. For still weaker competition it rises again slightly, to 0.176 at $r_0 = 1000$, although forage-fish biomass keeps increasing. Between $r_0 = 10$ and 1000 the number of adult forage fish (4–20 g) in State RF increases by only 15%, while their feeding level rises from 0.44 to 0.60. Because a predator kills prey in proportion to $1 - f$ (Eq. (10)), the better-fed forage fish eat fewer predator larvae each, and peak larval predation mortality falls to 36.8 year^{-1}.
2. *Competition reduces, but never reverses, the food benefit.* With State RF growth and State RFP mortality, $R_{0,\text{Pred}}$ exceeds its State RFP value $1/(1 - L_R) = 4.55$ at every r_0, rising from

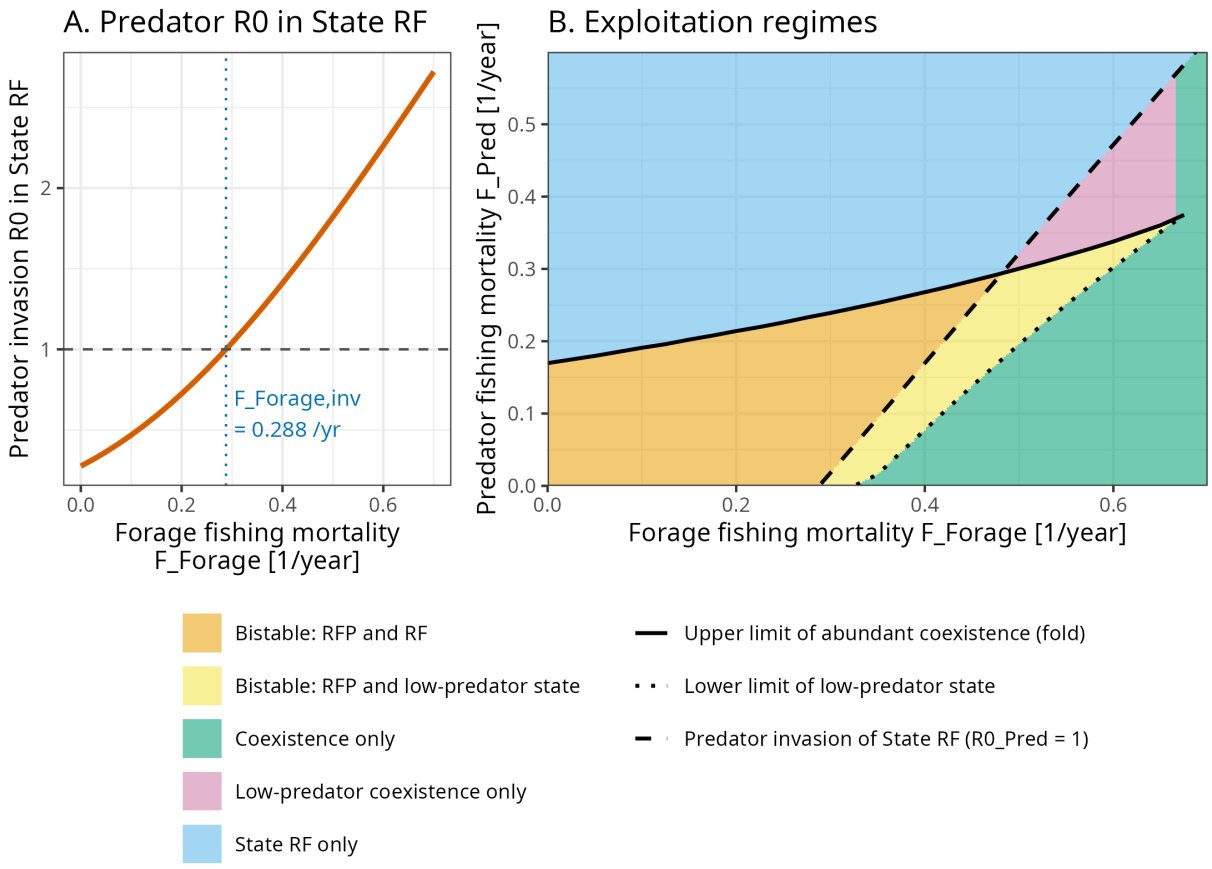


Figure 5: (A) Predator invasion reproduction number in State RF as a function of forage-fish fishing mortality; State RF becomes invasible above $F_{\mathrm{Forage,inv}}$. (B) Regimes in the $(F_{\mathrm{Forage}}, F_{\mathrm{Pred}})$ plane computed from the upper limit of the abundant coexistence state (solid), the lower limit of the depleted state (dotted) and the predator invasion boundary of State RF (dashed). Beyond the cusp the abundant and low-predator states form one continuous branch; the vertical edge of the pink region there marks where the branch at the upper limit falls below the 100 g used to tell the two apart, not a bifurcation.

6.32 at $r_0 = 0.25$ to 14.95 at $r_0 = 1000$: in State RF the forage fish always provide more food to adult predators than they take from predator juveniles. Competition is nonetheless visible in the juveniles. In State RF at $r_0 = 1$, predators of 1–50 g feed at a mean feeding level of 0.50 instead of 0.61 and reach maturation at an age of 1.53 instead of 1.34 years along the mean growth trajectory (Figure 6C); at $r_0 = 1000$ they feed at 0.64 and mature after 1.10 years, earlier than in State RFP.

Competition between forage fish and predator juveniles therefore never creates the invasion barrier. Its effect on the depensation is mainly indirect: by limiting the forage fish when the predator is absent, strong competition limits larval predation and opens State RF to invasion, while with weak competition satiated forage fish and faster juvenile growth weaken the barrier from the other side.

Invasion of State RF does not, however, mean recovery. For $r_0 < 0.595$ a predator reintroduced into State RF at 0.5% of its State RFP abundance settles at a low abundance (49.2 g at $r_0 = 0.25$ and 11.5 g at $r_0 = 0.5$, against 584 g in State RFP), as it does above $F_{\mathrm{Forage,inv}}$ in the reference model (Section 3.5). The unfished system is thus still bistable, between State RFP and a low-predator state instead of State RF. The predator can invade State RF up to $F_{\mathrm{Pred}} = 0.280$ year^{-1} at $r_0 = 0.25$, well above the collapse threshold, so overfishing leads to the low-predator state, from which a forage-fish harvest above $F_{\mathrm{Forage,rec}} = 0.151$ and 0.234 year^{-1} is still needed to restore the abundant state.

The fishing thresholds respond accordingly (Figure 6D). The collapse threshold F_{crit} increases monotonically with r_0, from 0.069 year^{-1} at $r_0 = 0.25$ to 0.208 year^{-1} at $r_0 = 1000$, and the collapse remains a jump from an abundant state (354–475 g of predators just below F_{crit}). Strong competition thus makes the abundant state more fragile, consistent with a second feedback: forage fish released by predator fishing also deplete the food of juvenile predators. The forage-harvest thresholds instead mirror $R_{0,\mathrm{Pred}}(\mathrm{RF})$: $F_{\mathrm{Forage,inv}}$ is largest (0.313 year^{-1}) at $r_0 = 5$ and $F_{\mathrm{Forage,rec}}$ (0.365 year^{-1}) at $r_0 = 2$, and both decrease towards stronger and weaker competition. State RFP is stable at every r_0 (leading eigenvalue between -0.312 and -0.016 year^{-1}, all real; for $r_0 \leq 2$ the slowest mode is the relaxation of the resource itself).

3.7 Robustness

Numerical resolution. With the second-order scheme and physical diffusion, the thresholds are converged: halving or doubling the number of size classes changes F_{crit} by at most 0.001 year^{-1} and $R_{0,\mathrm{Pred}}(\mathrm{RF})$ by at most 2% (Table 3). The original first-order discretisation without physical diffusion gives $F_{\mathrm{crit}} = 0.176$ and $R_{0,\mathrm{Pred}}(\mathrm{RF}) = 0.493$, against 0.170 and 0.276 in the reference model. Its numerical diffusion is of a similar magnitude to the physical diffusion assumed here, so its qualitative results stand, but its thresholds depend on the grid rather than on any biological assumption; its forage-harvest thresholds ($F_{\mathrm{Forage,inv}} = 0.119$, $F_{\mathrm{Forage,rec}} = 0.135$ year^{-1}) are less than half those of the reference model.

Size diffusion. Figure 7 shows the thresholds for size diffusion between $c = 0.02$ and $c = 0.3$. Over this range, $R_{0,\mathrm{Pred}}(\mathrm{RF})$ varies between 0.21 and 0.41, F_{crit} between 0.158 and 0.199 year^{-1}, $F_{\mathrm{Forage,inv}}$ between 0.249 and 0.292 year^{-1}, and $F_{\mathrm{Forage,rec}}$ between 0.323 and 0.337 year^{-1}. The predator cannot invade State RF at any of these diffusion levels, so the bistability and the failure of a moratorium do not depend on the assumed growth variability.

Calibration. Calibrating the reference model fully to the nominal State RFP targets of 100 g and 600 g, rather than stopping after the two rounds of the recipe, lowers $R_{0,\mathrm{Pred}}(\mathrm{RF})$ to 0.220 and F_{crit} to 0.153 year^{-1}, and raises the forage-harvest thresholds to $F_{\mathrm{Forage,inv}} = 0.340$ and $F_{\mathrm{Forage,rec}} = 0.394$

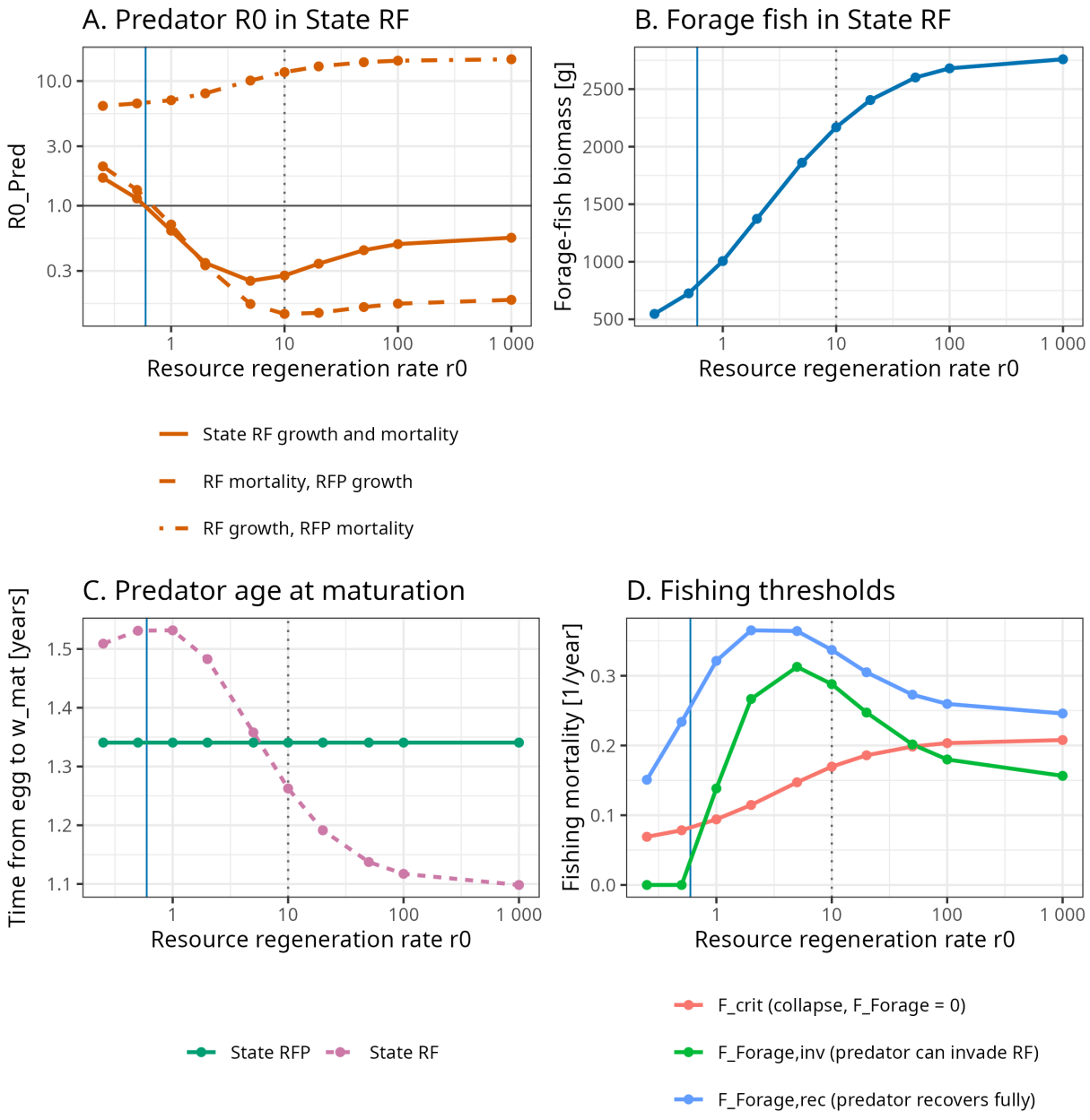


Figure 6: Dependence of the results on the strength of competition for the resource, varied through the resource regeneration rate r_0 [$g^{1/3}$ $year^{-1}$] with State RFP held fixed; small r_0 means strong competition. The dotted line marks the reference model, the thin blue line the value of r_0 at which $R_{0,\mathrm{Pred}}(\mathrm{RF}) = 1$; to its left the predator can invade State RF without fishing. (A) Predator invasion reproduction number in State RF (solid) and its decomposition into the effect of State RF mortality with State RFP growth (dashed) and of State RF growth with State RFP mortality (dot-dashed). (B) Forage-fish biomass in State RF. (C) Age at which a predator following the mean growth trajectory reaches maturation in States RFP and RF. (D) The collapse threshold without forage-fish harvest and the forage-fish mortalities above which the predator can invade State RF and recovers fully.

Table 3: Dependence of the results on numerical resolution, model structure and calibration. Each variant is rebuilt and calibrated until its State RFP biomasses and reproduction levels equal those of the reference model (relative error below 10^{-5}), except the last, which is calibrated in the same way to the nominal targets of 100 g and 600 g with $L_R = 0.65$ and 0.78. Bistability between RFP and RF without fishing requires $R_{0,\mathrm{Pred}}(\mathrm{RF}) < 1$. The reference model has 320 size classes. The original model is the first-order discretisation with 80 size classes and without physical diffusion used in the first version of this study. In the variant $w_R = 1$ g the resource tapers off at 1 g instead of 10 g. F in diet: percentage of the predator's food that comes from forage fish at State RFP.

Variant	$B_{\mathrm{For}}(\mathrm{RF})$	$R_{0,\mathrm{For}}(\mathrm{RP})$	$R_{0,\mathrm{Pred}}(\mathrm{RF})$	F_{crit}	$F_{\mathrm{Forage,inv}}$	$F_{\mathrm{Forage,rec}}$	F in diet
Reference	2169	2.41	0.276	0.170	0.288	0.337	20%
160 classes	2167	2.41	0.281	0.170	0.285	0.335	20%
640 classes	2169	2.41	0.274	0.169	0.289	0.338	20%
Original	2187	2.55	0.493	0.176	0.119	0.135	19%
$w_R = 1$ g	3221	10.34	0.144	0.467	0.329	0.675	37%
Nominal targets	2266	2.37	0.220	0.153	0.340	0.394	21%

year^{-1} (Table 3). The qualitative picture is unchanged, but differences of a few per cent in the calibrated biomasses move the fishing thresholds by up to 18%.

Resource size range. Moving the resource taper from 10 g to 1 g, so that adult predators depend more on forage fish (37% of their diet instead of 20%), strengthens the depensation: $R_{0,\mathrm{Pred}}(\mathrm{RF})$ falls to 0.144 and F_{crit} rises to 0.467 year^{-1}, and the predator recovers fully only above $F_{\mathrm{Forage}} = 0.675$ year^{-1}.

The result is, however, sensitive to cannibalism in either species (Figure 8). $R_{0,\mathrm{Pred}}(\mathrm{RF})$ increases steeply with forage-fish cannibalism and exceeds one at $\theta_{\mathrm{For,For}} = 0.120$. When forage fish eat their own larvae, they regulate their own recruitment, State RF contains fewer forage fish (1464 g instead of 2169 g at $\theta_{\mathrm{For,For}} = 0.15$), and the predator can invade. Cultivation–depensation in this model therefore requires that adult forage fish prey on predator larvae at least 8 times more readily than on their own larvae.

Predator cannibalism has the same qualitative effect, but through a different route: $R_{0,\mathrm{Pred}}(\mathrm{RF})$ exceeds one at $\theta_{\mathrm{Pred,Pred}} = 0.360$. Because the predator is absent from State RF, $\theta_{\mathrm{Pred,Pred}}$ does not enter the invasion calculation directly (Eq. (10) multiplies it by the predator density); changing it in the reference model without recalibrating leaves $R_{0,\mathrm{Pred}}(\mathrm{RF})$ unchanged. Its effect arises entirely through the calibration. To sustain the same State RFP despite cannibalism, the calibration must raise the predator's reproductive efficiency, from $\epsilon_{\mathrm{Pred}} = 3.0 \times 10^{-5}$ at $\theta_{\mathrm{Pred,Pred}} = 0.05$ to 1.5×10^{-4} at 0.4, while State RF itself hardly changes (2169–2435 g of forage fish across the scan). A predator invading State RF keeps this high reproductive efficiency but has no conspecific adults to eat its larvae.

The predator-cannibalism threshold depends on how the variants are calibrated. In the first version of this study each variant received only the fixed recipe of Section 2.2, which falls increasingly short of the targets as cannibalism increases: State RFP then held 271–584 g of predators and 54–94 g of forage fish across the scan, so that the resource capacity and State RF changed as well, and the threshold was $\theta_{\mathrm{Pred,Pred}} = 0.149$ (grey in Figure 8). With every variant calibrated to the

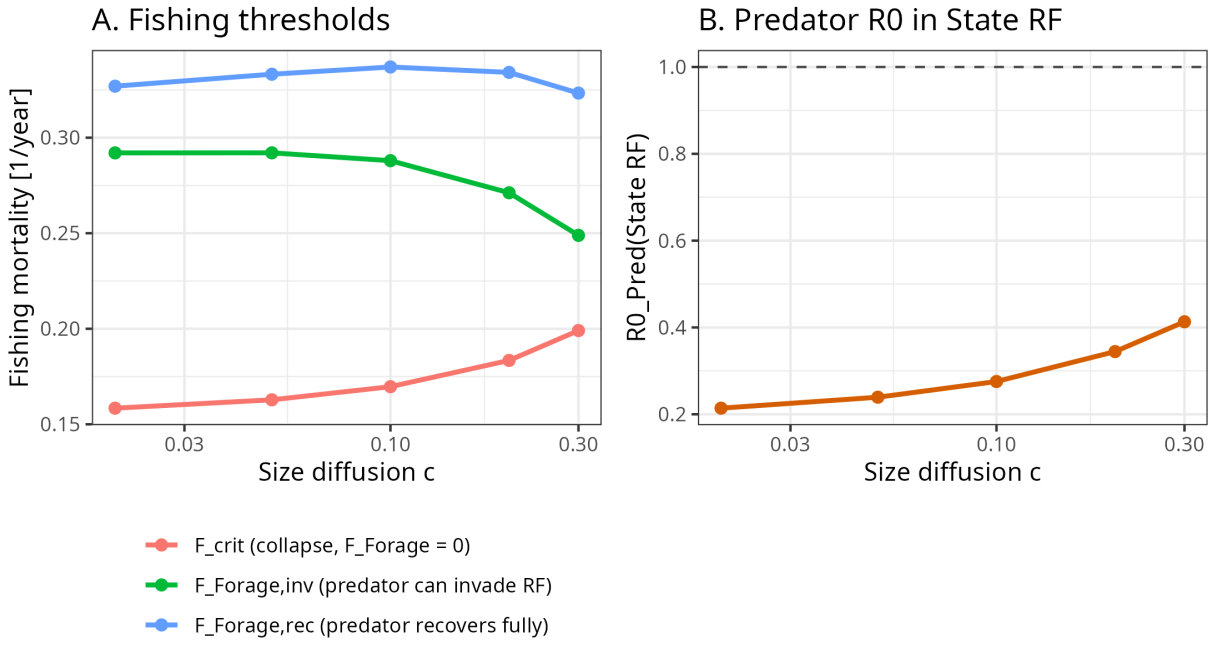


Figure 7: Dependence of the fishing thresholds (A) and of the predator invasion reproduction number in State RF (B) on the strength of size diffusion $c = d\sigma^2/d\ln w$. Each point is a separately calibrated model with the State RFP biomasses and reproduction levels of the reference model, which has $c = 0.1$. For $c = 0.02$ the van Leer limiter replaces the centred flux.

reference State RFP, the threshold is more than twice as large. For forage-fish cannibalism, where the recipe nearly reaches the targets, the two protocols agree closely (0.118 and 0.120). In short, cultivation–depensation arises here only if forage fish, rather than the predator's own adults, are the main predators of predator larvae; predator cannibalism destroys it only when it reaches about 36% of the rate at which predators eat forage fish.

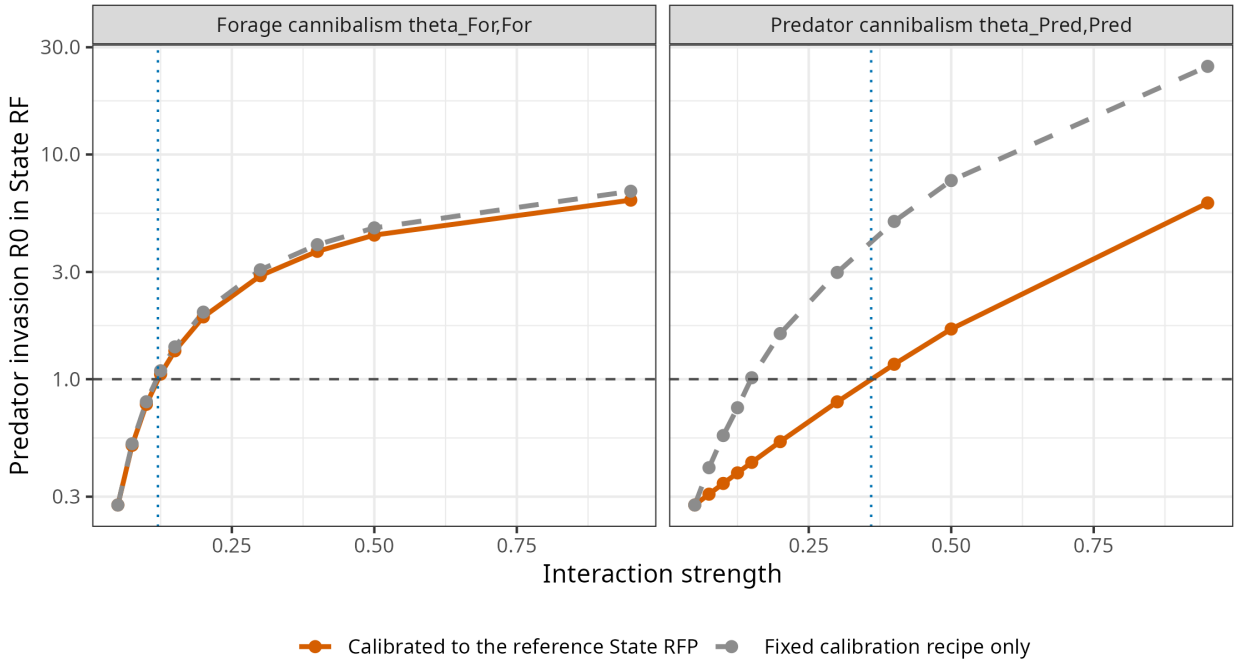


Figure 8: Predator invasion reproduction number in State RF as a function of the strength of forage-fish cannibalism $\theta_{\mathrm{For,For}}$ (left) and predator cannibalism $\theta_{\mathrm{Pred,Pred}}$ (right), with all other entries of the interaction matrix unchanged. Each point is a separately calibrated model: orange, calibrated until State RFP has the biomasses and reproduction levels of the reference model (the dotted line marks the root of $R_{0,\mathrm{Pred}}(\mathrm{RF}) = 1$); grey, calibrated with the fixed recipe alone, as in the first version of this study, which leaves State RFP increasingly far from the reference as predator cannibalism increases. To the right of the dotted line the predator can invade State RF and there is no bistability without fishing.

4 Discussion

4.1 Size-Structured Ontogeny and Cultivation–Depensation

Cultivation–depensation emerges in the standard size-spectrum dynamics without any explicitly depensatory stock–recruitment function: a forage fish and a predator that spawn at the same size and interact through size-based feeding can produce alternative stable states, a predator Allee effect and a moratorium that fails. The decomposition of $R_{0,\mathrm{Pred}}$ shows that the mechanism here is predation on predator larvae, as in the original hypothesis (Walters and Kitchell 2001), and not the food-dependent development that underlies emergent Allee effects in stage-structured consumer–resource models (de Roos and Persson 2002, 2013; van Kooten, de Roos, and Persson 2005). Indeed the food effect works against depensation: an uncultivated system with many forage fish would

be a rich feeding ground for the few predators that survive to adulthood. Distinguishing these mechanisms in real systems is difficult, because both produce the same population-level signature (Gårdmark et al. 2015).

Competition for the resource, the second route in the original hypothesis (Walters and Kitchell 2001), plays a different role. Juvenile predators do suffer from competition with forage fish in State RF when the resource regenerates slowly, but at no strength of competition does this outweigh the extra food that the forage fish provide to adult predators (Section 3.6). The main effect of competition is on the forage fish themselves: competition among forage fish caps their abundance once the predator is gone, and so caps the larval predation that maintains the depensation. Strong competition therefore weakens the depensation, and below a regeneration rate of $r_0 \approx 0.60$ g$^{1/3}$ year^{-1} the predator can invade State RF. A reintroduced predator still settles at low abundance there, so the alternative state survives in a weaker form, and a forage-fish harvest is still needed for full recovery. Without competition the depensation also weakens, because well-fed forage fish eat fewer larvae each, but it persists: even when the fish never deplete the resource by more than 1.5% ($r_0 = 1000$), $R_{0,\mathrm{Pred}}(\mathrm{RF}) = 0.55$. The barrier is strongest at intermediate competition, which shows that the strength of resource competition is not a detail but one of the parameters that decide whether cultivation–depensation occurs.

The mechanism is, however, conditional. It requires that forage fish prey on predator larvae much more than on their own larvae, and that predator larvae are not mainly eaten by their own adults. With symmetric larval predation, the self-regulation of the forage fish keeps State RF sparse enough for the predator to invade; with strong predator cannibalism, a predator invading State RF escapes the main source of its larval mortality. A biological basis for the asymmetry must come from spatial or seasonal segregation of spawning and feeding grounds, which a single well-mixed size spectrum cannot represent. Cod, the motivating example, are notably cannibalistic, so the second condition in particular deserves empirical scrutiny. In addition, the predator in this model obtains most of its food from the resource, which extends beyond 10 g; restricting the resource to small sizes strengthened rather than weakened the depensation, but the model's "predator" is closer to a generalist than to an obligate piscivore.

4.2 Priority Effects

The invasion numbers establish a priority effect: the full food web assembles if the predator arrives first but not if the forage fish does. This is the pattern expected from life-history intraguild predation, in which each species preys on the other's young (Holt and Polis 1997; Hin et al. 2011). The forage fish, which preys on the predator's young, can exclude the predator; the predator, whose adults eat forage fish but cannot eliminate them, cannot exclude the forage fish.

4.3 Implications for Exploitation

The model supports three management conclusions, each with qualifications.

1. **Selective predator fishing risks an irreversible collapse.** Without forage harvest, the abundant predator state is lost at a fold at $F_{\mathrm{Pred}} = 0.170$ year^{-1}, and a predator moratorium alone cannot restore it because State RF is uninvasible. Whether real systems are locked in this way is an empirical question; the Scotian Shelf, where the forage-fish outbreak eventually collapsed and cod began to recover (Frank et al. 2011), shows that additional dynamics not in this model, such as the forage fish overshooting its own food supply, can open an escape route.

2. **Forage-fish harvesting raises the collapse threshold.** At $F_{\text{Forage}} = 0.35$ year^{-1} the threshold increases by 49%. The collapse remains abrupt, although above $F_{\text{Forage}} \approx 0.48$ year^{-1}, where the invasion boundary of State RF lies above the fold, it leads to a low-abundance predator state rather than to extinction; only beyond the cusp at $F_{\text{Forage}} \approx 0.650$–$0.675$ year^{-1} does the predator decline continuously.
3. **Forage-fish harvesting can restore a collapsed predator.** Once $F_{\text{Forage}} > 0.288$ year^{-1}, a reintroduced or immigrating predator can invade, and once $F_{\text{Forage}} > 0.337$ year^{-1} it recovers fully, re-establishes top-down control, and the community stays in State RFP when the forage harvest ends. The difference between the two thresholds matters for management: a forage harvest that just makes invasion possible leaves the predator as a remnant population. This is the model analogue of the whole-lake experiment in which culling of a prey and competitor population allowed a predator to recover (Persson et al. 2007), and of the recovery of predators in marine reserves through suppression of their competitors and predators (Baskett, Yoklavich, and Love 2006).

These conclusions concern fishing two species at different rates. They do not amount to *balanced harvesting* in the sense of Garcia et al. (2012), which distributes fishing mortality across all species and sizes in proportion to their productivity, and which has been analysed in size-spectrum models by Law, Plank, and Kolding (2012), Jacobsen, Gislason, and Andersen (2014) and Law, Plank, and Kolding (2016). Our forage-fish gear also takes juveniles from 1 g. The historical tendency of fisheries to move from large predators to lower trophic levels (Pauly et al. 1998; Essington, Beaudreau, and Wiedenmann 2006) is the opposite of the order that the model suggests would protect predators, but the model provides no estimate of the yields or ecosystem consequences of such a strategy.

4.4 Limitations

The results are for one hand-tuned parameter set. Calibrated quantities such as the predator's reproductive efficiency ($\epsilon_{\text{Pred}} = 3.0 \times 10^{-5}$) directly scale its invasion number, and the target biomasses are illustrative. The reference model is defined by a calibration recipe that stops short of these targets; calibrating it fully to 100 g and 600 g changes $R_{0,\text{Pred}}(\text{RF})$ from 0.276 to 0.220, F_{crit} from 0.170 to 0.153 year^{-1}, and $F_{\text{Forage,inv}}$ and $F_{\text{Forage,rec}}$ from 0.288 and 0.337 to 0.340 and 0.394 year^{-1} (Table 3). The thresholds should therefore be read as properties of this particular parameter set, not as estimates. The resource regeneration rate is also a free parameter, and the reference value lies close to the one at which the invasion barrier is strongest (Section 3.6); for both stronger and weaker competition for the resource the forage-harvest thresholds are lower. The model is deterministic, spatially unstructured and has a single resource spectrum; recovery requires a reintroduced or immigrating seed; and each gear catches only one species. The thresholds depend quantitatively on the assumed size diffusion (Figure 7), which is poorly constrained, and on the senescence mortality that limits how far adults diffuse beyond w_∞, whose form is an assumption.

4.5 Reproducibility

All model construction, calibration, steady-state and stability analysis, bifurcation mapping, robustness checks and figures are produced by `cultivation_depensation.R` with `mizer` 3.4.0.9001. The script requires `mizer` $\geq$ 3.4.0.9001: version 3.4 introduced the second-order scheme and the scheme-consistent invasion numbers, 3.4.0.9000 made the predation mortality second-order accurate (mizer issue 619), and 3.4.0.9001 stopped the encounter and predation integrals from wrapping around the size grid (mizer issue 618). Version 3.4.0.9001 is not yet released: the results were computed with `mizer`

built from commit `e1e85b0` of https://github.com/sizespectrum/mizer (the branch of pull request 627), which can be installed with `remotes::install_github("sizespectrum/mizer@e1e85b0")`; `results/session_info.txt` records the R session.

The script saves the reference model to `cultivation_depensation_params.rds` (written with `saveParams()`, to be read with `readParams()` in the same or a later `mizer` version) and every number quoted in this paper to `results/cultivation_results.rds`, which this document reads when it is knitted. The calibration of the reference model is the deterministic recipe in `build_cultivation_model()` (Section 2.2); a clean run reproduces it without any choice of stopping point. Every calibrated variant of Section 3.7 is saved with its resource capacity and numerical settings to `results/models/` (variants calibrated with the recipe alone, for comparison, to `results/models/recipe/`), and the point-wise values behind Table 3 and Figures 7 and 8, together with each variant's ϵ_i, $R_{\max,i}$, $L_{R,i}$, number of calibration rounds, final biomass error and biomass drift, are written to `results/robustness.csv`, `results/diffusion_scan.csv` and `results/cannibalism_scan.csv`. The variants of Section 3.6 differ from the reference model only in the resource regeneration rate and are not saved separately; their values are written to `results/competition_scan.csv`. The full run takes about three hours on four cores, of which the robustness checks take about an hour; `Rscript cultivation_depensation.R --robustness` repeats only the robustness checks, `--competition` only the scan of the resource regeneration rate, and `--figures` only redraws the figures.

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